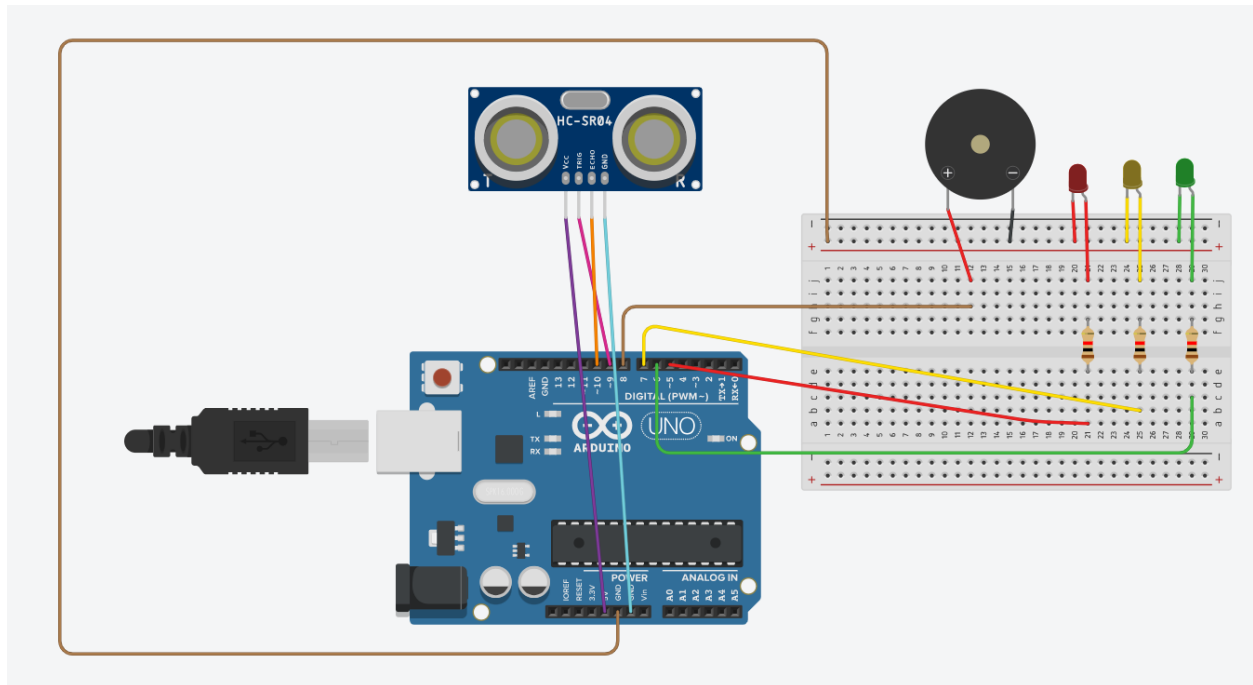


Intelligent Proximity Warning System using HC-SR04



Project Description

This project presents an **Intelligent Proximity Warning System** built using an Arduino and the HC-SR04 ultrasonic sensor. The system is designed to measure the distance between an object and the sensor in real time and provide visual and audio alerts based on predefined distance zones.

The system is divided into three zones:

- **Safe Zone (>30 cm):** Green LED glows indicating no immediate obstacle.
- **Warning Zone (11–30 cm):** Yellow LED glows and the buzzer beeps intermittently.
- **Danger Zone (≤ 10 cm):** Red LED glows and the buzzer produces a continuous sound alert.

The HC-SR04 ultrasonic sensor calculates distance by sending ultrasonic waves and measuring the time taken for the echo to return. Based on this distance, the Arduino processes the data and activates the corresponding LEDs and buzzer alerts.

Objectives

- To design a real-time distance monitoring system
- To implement multi-level alert mechanisms using LEDs and buzzer
- To demonstrate practical application of ultrasonic sensors

Components Used

- Arduino Uno-1
- HC-SR04 Ultrasonic Sensor-1
- LEDs (Red, Yellow, Green)-each 1
- Buzzer-1
- Resistors (1kohms)-3
- Jumper wires-req.

Working Principle

The ultrasonic sensor emits sound waves and calculates the distance using the time-of-flight principle. The Arduino continuously reads this distance and compares it with predefined thresholds to determine the alert level.

Applications

- Obstacle detection systems
- Parking assistance systems
- Robotics navigation
- Safety alert systems

Conclusion

This project demonstrates an efficient and low-cost method for proximity detection and alert systems. It can be further enhanced by integrating displays, IoT features, or mobile notifications for advanced applications.

Simulation in Tinker-Cad:

<https://www.tinkercad.com/things/9F4rhT9gjb4-intelligent-proximity-warning-system-using-hc-sr04>